

Sail Car

(Coach Scheduled Event - 45 Minutes)



Description: Teams will design, build, and race a sail-powered model using wind as their only source of energy, to travel as straight a distance as possible in the shortest amount of time. They will answer questions on a written assessment on their understanding of wind, angles, and aerodynamics.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Safety Equipment - Teams Must Bring:

- **Safety glasses labeled ANSI Z87+ (impact rated)**
- **All competitors must wear their eye protection during any competitor's test phase of the competition.**
- **If a team does not have the required eye protection, they will be given the opportunity to obtain it, time allowing, but will not receive extra time.**
- **If a team is unable to obtain eye protection, the team will not compete and will receive a no-show score.**

Impact Safety Glasses Required



Eyeglasses are not safety rated!

Teams must also bring:

- Writing Utensils
- a ruler or straight edge
- one pair of scissors

Materials Provided at the event:

- 4 ≈ 4 cm (1.5 inches) plastic wheels
- 6 ≈ 20 cm (8 inches) paper drinking straws
- 2 ≈ 7.5 cm (3 inches) cooking skewers
- 2 ≈ 15 cm (6 inches) cooking skewers
- 4 ≈ 15 cm (6 inches) craft sticks
- 1 ≈ 8.5 X 11-inch (21.6 cm x 28 cm) tissue paper
- 1 ≈ 8.5 X 11-inch (21.6 cm x 28 cm) 60–90-pound card stock paper
- 1 ≈ 13 cm by 7.5 cm (5 in x 6 in) piece of cardboard
- 1 meter of 2.54 cm (1 in) masking tape
- 10 grams (0.4 oz) of plasticine

THE COMPETITION

The Written Assessment: The team members will take a written assessment on their understanding of

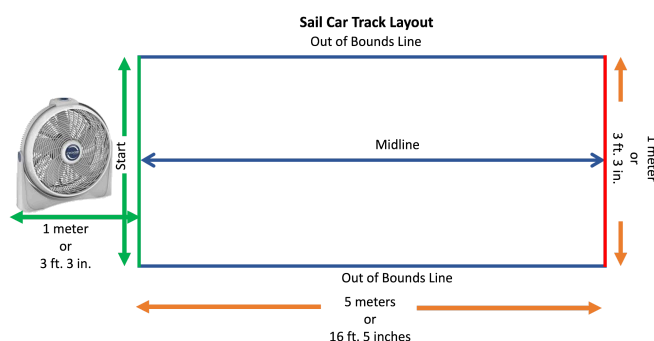
wind, angles, and aerodynamics. The test will take place simultaneously during the construction of the team car.

Construction Phase (30 minutes):

- Teams may cut their materials to any size they deem appropriate, except for the craft sticks which must remain whole
- Teams may use as much or as little of their materials to build their car.
- Teams may have 2 untimed practice runs to test their car within the 30 minutes build time with permission of the Event Supervisor.
- Practice runs will not occur after the 30-minute build time.
- A request for additional materials results in a team being placed in Tier 2. This means that the team may still participate but will be placed below all teams that do not violate this rule
- Releasing a car down the track without permission of the Event Supervisor during both the testing and scoring phases of the competition will result in a team being placed in Tier 2.

The Track:

- The track will be on a relatively smooth, flat, hard surface.
- The track will be 3 meters long, with a width of one meter.
- It will have a start line, end line and a center line.



Official Testing:

- Each team will have one officially scored run.
- The team will prepare and hold their car anywhere behind the start line.
- The team must state that they are ready to turn on the fan and must wait for the Event Supervisor to acknowledge they are ready to time
- At this signal a team member will make a loud announcement of, “3, 2, 1, LAUNCH!” and turn the fan on.
- The Event Supervisor will record the time until the team
- Teams that push their car at any time will be placed in Tier 2.

SCORING:

Final Score (FS) = Assessment Score + Track Score, max score = 100 points

- Assessment score = $(TAS \div CAS) \times 60$; rounded to the 100th place value; maximum 50 points
 - Team Assessment Score (TAS) = team’s score on the assessment
 - Competition Assessment Score (CAS) = the highest assessment score for all teams
- Track Score = $(CTS \div TTS) \times 40$; rounded to the 100th place value; maximum 50 points
 - Competition Track Score (CTS) = the lowest Track Score for all teams
 - Team Track Score (TTS) = the team’s Track Score
 - Each Track Score = Time + Distance. Want the Track Score to be as low as possible.
 - Distance
 - If the sail car crosses the finish line in bounds, this is the distance (in cm) along the finish line from the midline to where the car crossed the finish line
 - If the sail car crosses out of bounds before the finish line, this is the straight line distance (in cm) from where the vehicle went out of bounds to the midline of the finish line.

- If the sail car stops before the finish line, but remains in bounds, this is the straight line distance (in cm) from the front edge of the car to the midline of the finish line.
 - Time for the Sail Car's run. Timing ends when the sail car passes the finish line, goes out of bounds, or stops within bounds.
- Tiers
 - Tier 1 - the sail car is not in Tier 2
 - Tier 2 (the lower scoring tier)
 - The team requests additional build materials.
 - The team takes a disallowed practice run.
 - The team pushes their car during their official test.

Scoring Example:

- Competition Assessment Score (CAS) = 15. The team scores 9 on the assessment.
Assessment Score for the team= $(9 \div 15) \times 60 = 36$ points
- Competition Track Score (CTS) = 1.45 seconds + 0 cm = 1.45 (sail car crosses finish at the midline)
The Team's Track Score (TTS) = 2.56 seconds + 1 cm = 3.56 (sail car stops just short of the finish)
Track Score for the team= $(1.45 \div 3.56) \times 40 = 17.79$ points
- Final Score (FS) = $36 + 17.79 = 53.79$ points

Tiebreakers:

- 1st - The team closest to the midline when passing the end line
- 2nd - The better score on the written test
- Select questions on the test

Resources:

- [NASA - What Is Aerodynamics?](#)
- [Kid Wind – Sail Cars Activity](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles